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A low-latency intelligent edge computing architecture for real-time decision-making in large-scale IoT networks

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Abstract
The deployment of massive Internet of Things (IoT) networks has further heightened demand for real-time analytics, autonomous decision-making, and ultra-low-latency processing. Traditional cloud-centric systems suffer from delays in communication, network overload, and a lack of flexibility in dynamic systems such as industrial automation, smart healthcare, and connected mobility. The main problem is to achieve scalable, smart, distributed processing while remaining reliable, efficient, and coordinating across heterogeneous nodes in the IoT. The paper aims to solve this issue by introducing ELARA (Edge-Level Adaptive Reasoning Architecture)- a hybrid intelligent infrastructure that will maximize the placement of computation and accelerate real-time inference. The methodology incorporates: (i) Federated Feature Extraction to reduce transmission of raw data, (ii) Adaptive Task Offloading Engine based on workload-sensitive scheduling, and (iii) Reinforcement-based Resource Orchestration based on predictive scaling, elasticity, and edge-node collaboration. Also, ELARA uses a consensus-based micro-cluster architecture to support fault tolerance and distributed learning without a centralized cloud. Experimental results indicate that ELARA delivers 39–52 ms end-to-end latency, saves up to 48% Bandwidth, 31% energy, and 850–932 samples/sec inferences, which is better than current models, including CEA, DSS, and DFJSP. It also provides higher task completion rates (93–98%), higher offloading accuracy (89–93%), and much faster recovery from failures. To sum up, ELARA creates a scalable framework for intelligent, low-latency, and robust edge-based IoT systems that can be used in emerging autonomous applications.

Keywords Edge computing, IoT, Federated intelligence, Reinforcement learning, Real-time systems, Distributed architecture

1 Introduction

The recent increase in the scale of large-scale IoT networks has introduced an unparalleled need to deploy smart [1]. Distributed computing systems with a proven ability to operate real-time streams of data with low latency [2]. Traditional cloud-based models are efficient for large-scale storage and more complex analytics [3]. Introduces

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significant communication overhead and latency, and therefore, cannot be used in time-sensitive applications such as smart healthcare monitoring, industrial automation, and automated transportation [4]. With increased density and heterogeneity in IoT deployments, it has become critical to enable faster, reliable, and context-aware decision-making at the network edge to keep systems operating more responsively [5].

Low-latency edge intelligence is one of the core necessities of next-generation IoT infrastructures [2]. Current architectures are usually unable to balance computational load, offer flexibility during dynamic workloads, and efficiently utilize resources across geographically distributed edge nodes. These limitations impair the operation of latency-sensitive IoT applications and limit scalability in real-world settings [6].

To address these constraints, this paper proposes ELARA, a hybrid intelligent computing platform [7]. That can improve real-time inference by optimizing computation placement and distributed learning. ELARA combines federated feature extraction, workload-aware adaptive offloading, and reinforcement-based resource orchestration to make decisions faster and with minimal communication overhead [8]. Using a consensus-based micro-cluster architecture guarantees fault tolerance and elastic scaling [9]. In general, ELARA offers a solid baseline for developing efficient, resilient, and intelligent edge ecosystems of emerging autonomous IoT applications [10].

A. Low-latency edge intelligence

The growing size of IoT networks is necessitating the need to make decisions quickly based on immediate context at the network edge [11]. Low-latency edge intelligence also enables sensors, devices, and microcontrollers to handle data locally, reducing reliance on remote cloud servers [12]. This needs to change in systems such as autonomous systems, industrial robotics, and smart healthcare, where the performance or safety of the system can be compromised by a few milliseconds of latency [13]. The use of lightweight inference models and distributed processing would ensure real-time responsiveness, high reliability, and the continued exploitation of data streams in dynamic environments through edge intelligence [14].

B. Cloud-edge limitations

Conventional cloud-based solutions are highly limited in accommodating large-scale IoT systems [15]. Real-time analytics can be impaired by high communication latency, bandwidth overload, and unreliable connectivity, which are not suitable for mission-critical streams [16]. The variability of the workload, the inability to scale, and inefficient resource utilization across heterogeneous nodes are issues with the application of fixed-edge systems as well. The weaknesses lead to delays in decision-making, inaccuracy, and excessive overhead in the operations [17]. To avoid computation cycling and edge reconfiguration, and to ensure smooth inter-edge synchronization, are necessary to sustain performance under varying conditions.

C. ELARA motivation

The ELARA framework is presented to solve the underlying limitations of existing IoT and edge-computing frameworks. It focuses on adaptive reasoning, task placement optimization, and distributed learning to improve real-time inference in large networks. ELARA minimizes communication delay and effectively utilizes resources by

incorporating federated feature extraction, dynamically offloading tasks, and task reinforcement. Its micro-cluster consensus design makes it resilient, fault-tolerant, and scalable for decision-making. ELARA is motivated by the need to support powerful, low-latency, and intelligent computation for next-generation autonomous IoT applications.

ELARA is a fundamentally different edge intelligence system that combines micro-cluster coordination via consensus with adaptive offloading via reinforcement in a fully decentralized system. In contrast to current edge-fog systems, which use either static tiered control, heuristic scheduling, or centralized fog coordination, ELARA enables autonomous, peer-level coordination of edge nodes within micro-clusters, eliminating single points of control and enhancing fault tolerance. Moreover, whereas past methodologies typically adopt load-aware or rule-based offloading, ELARA incorporates reinforcement-based resource orchestration that continuously learns superior task placement and scaling policies in response to dynamically changing workloads. The federated feature extraction also makes ELARA unique, as it doesn't transmit large volumes of raw data while still maintaining inference accuracy. All these design decisions enable ELARA to achieve lower latency, better scalability, faster failure recovery, and improved resource utilization than the current strategies, including CEA, DSS, and DFJSP.

2 Related works

The latest technologies in IoT, AI, edge-fog computing, reinforcement learning, and cyber-physical systems have revolutionized decision-making, resource optimization, security, and automation across industries. The analyzed papers unanimously point to new frameworks, optimization tools, and intelligent architectures that will improve the scalability, performance, accuracy, and real-time responsiveness of intelligent environments, industrial systems, energy networks, and agriculture.

IoT needs to support real-time decision-making, which means scheduling tasks efficiently under constraints on data freshness and latency. New strategies present prioritization models and adaptive schedulers to optimize execution timelines. It focuses on scalable architectures that can support heterogeneous IoT workloads. The suggested cutting-edge approaches (CEA) method maximizes timing and delivers high-quality decision outputs in dynamic IoT ecosystems [18].

Interconnected Intelligence integrates IoT, Fog, and Cloud to reduce latency and improve resource utilization. The hybrid multi-layer architecture is a distributed architecture that allocates computation to devices at the Fog and Cloud levels. Under varying workloads, experimental validation exhibits better accuracy, energy savings, and load balancing. The combined HA strategy empowers real-time decision-making in industrial automation and smart urban settings [19].

Mass IoT implementations generate high-volume data that requires fast analytics. Big data structures, especially Spark streaming, facilitate real-time predictions and scalable processing. The integration of machine learning makes the extraction of insights promptly. The proposed Spark-ML architecture is useful for supporting weather-station-type situations, where the processing throughput and responsiveness of live IoT monitoring are increased [20].

The IoT industrial environment requires machine-to-machine connectivity with low packet loss and throughput. Smart factories have the advantage of adaptive communication systems that remain stable as traffic increases. The NF-based model of

communication enhances the accuracy of autonomous decision-making and ensures a stable output production and high reliability of dynamic industrial processes [21].

Decision-making patterns that use less energy ensure that increasing SLAs, data centers, and user requests make up an ever-growing IoT ecosystem. Multi-level taxonomies are used in Decision Support Systems (DSS) to categorize energy-conscious mechanisms of smart cities and industrial systems. The energy-saving strategies of the suggested DSS plans demonstrate emerging trends in energy conservation and enhance the capabilities of urban intelligent computers [22].

Cyber-physical production systems are based on interoperable IoT-driven logistics and AI-based process analytics. Massive surveying emphasizes the precision of real-time sensor networks and production control with the assistance of machine learning. The importance of data-driven optimization is quantitatively evaluated in 164 studies. Improved CPPS systems improve smarter scheduling and sustainable industrial processes [23].

Edge computing is used to distribute computing capabilities closer to IoT devices, thereby reducing latency and improving human-computer interaction. An SDN-NFV combined architecture enhances memory, storage, and CPU performance and reduces network delays. The optimization framework based on the Deep Q-Network (DQN) is much more efficient in resource allocation and enhances user-perceived system usability in real-time IoT settings [24].

The dynamics of job arrival in manufacturing make scheduling a number of machines difficult. Multi-action decision problems in real-time problems are solved using hierarchical reinforcement learning. The GNN-MLP combined design allows proper sequencing and machine placement. The proposed dynamic flexible job shop scheduling problem (DFJSP) model also exhibits high-quality scheduling performance compared to metaheuristics at the industry level [25].

Federated Learning enables privacy-preserving analytics in distributed healthcare IoT systems. Training models at Edge, Fog, and Cloud levels will enable systems to adapt to heterogeneous medical data and minimize communication overhead. The optimization of energy consumption, latency, cost, and execution time of the proposed FedSDM framework is achieved by preferring edge-based deployments for real-time ECG monitoring applications [26].

IoT infrastructures are prone to high levels of security risks, most notably due to evolving DDoS attacks targeting the network and application layers. The intrusion detection plans use multi-level data detection to increase the resilience of heterogeneous devices. The suggested models for IoT-DDoS IDS enhance threat detection and facilitate the safe implementation of IoT technologies in open, industrial, and personal settings [27].

Deep reinforcement learning enables scalable resource allocation across various IoT networks online. New algorithms address mixed-action spaces and multi-agent interaction under severe QoS constraints. The suggested DRA-DR architecture offers adaptive scheduling and optimal resource distribution, allowing real-life IoT systems to cope with both complex and highly dynamic operational conditions [28].

Intelligent control strategies have optimized the energy systems of autonomous hydrogen production systems powered by renewable energy sources. An off-grid H2-RES design combines data-driven decision-making and mathematical modeling to minimize the production cost and increase safety. The energy scheduling policy using DDPG is

better than PSO and DQN because it maximizes the use of renewable sources, thereby increasing production efficiency and dispatch stability in large-scale hydrogen generation settings [29].

The AI-IoT combination is transforming precision agriculture by enabling remote sensing, high-throughput phenotyping, and autonomous robots that deliver real-time crop analytics. Case studies show how automation can enhance monitoring of fruit load, soil conditions, and plant traits. The presented AI-IoT Precision Agriculture framework focuses on scalable sensing, robust segmentation, and connectivity improvements, which address significant gaps in dataset integration, real-time decision-making, and usability at the farm level [30].

This motivation aligns with the paradigm of fog edge computing, explicitly moving latency-sensitive analytics and control nearer to data sources to address cloud-centric communication delays, bandwidth bottlenecks, and scalability constraints in large IoT systems, as formally defined by Hassan et al. (2025) [31], who characterize the change in paradigm of cloud computing to fog computing as a required architectural revolution of real-time and geographically distributed applications.

ELARA uses a decentralized micro-cluster-based coordination model, rather than the hierarchical coordination scheme of the multi-tier vehicular edge-fog architecture proposed by Hassan et al. (2025) [31], which entails centralized traffic optimization via hierarchical links among vehicles, roadside edge nodes, and fog servers. Higher workloads in the framework of Hassan et al. ensure that workload and decision-making are handled at higher levels of the mogs, which may create coordination overhead and scale limitations in the presence of dense, highly dynamic workloads. By contrast, ELARA focuses on consensus-based edge micro-clusters that enable peer-to-peer cooperation, local fault tolerance, and autonomous task redistribution without constant contact with higher-level fog or cloud controllers.

ELARA builds upon this research by incorporating reinforcement learning into adaptive offloading and resource control, enabling each edge/fog agent to learn the best offloading action over time from state transitions and performance feedback, rather than relying on heuristic thresholds or fixed policies. This reinforcement control enables ELARA to achieve more responsive, context-sensitive resource allocation in highly dynamic environments, especially when workload patterns are highly variable in both time and space.

2.1 Research gap

Despite the evolution of IoT, AI, and edge-fog-cloud decision-making, several limitations remain. The majority of research is not integrated into a holistic, real-time approach; instead, the separate tasks are treated as a system of intelligence. Numerous frameworks disregard security, privacy, and scalability, particularly in heterogeneous, large-scale deployments. The topics of energy efficiency and resource optimization are not consistently covered across different environments. Several solutions are computationally intensive, making them impractical to use. Also, the validation reality is limited to a few studies that use simulations as the basis, rather than actual IoT implementation, as shown in Table 1.

The innovations that exist include real-time IoT decision systems, adaptive resource scheduling, intelligent energy management, industrial automation, federated healthcare

Table 1 Summary of related works

Reference	Environment	Algorithm/framework	Real-time decision	Security/privacy	Resource efficiency
[18]	IoT	Real-time decision scheduling (CEA)	✓	X	Medium
[19]	IoT + Fog + Cloud	Hybrid architecture (HA)	✓	✓	High
[20]	IoT + Big data	Spark ML streaming	✓	X	Medium
[21]	IIoT	NF-based communication system	✓	X	High
[22]	IoT + DSS	Energy-aware DSS	✓	✓	Medium
[23]	CPPS/IoT	AI-driven CPPS analytics	✓	X	Low
[24]	Edge + SDN + NFV	DQN-based resource allocation	✓	X	High
[25]	Smart manufacturing	Hierarchical RL (HRL)	✓	X	High
[26]	Edge–Fog–Cloud	FedSDM (Federated smart decision making)	✓	✓	Very High
[27]	IoT Security	DDoS detection techniques	X	✓	Low
[28]	IoT	DRL-based resource allocation	✓	X	High
[29]	Renewable energy system	DDPG-based H_2-RES scheduling	✓	✓	High
[30]	AI + IoT agriculture	AI–IoT precision agriculture	✓	X	Medium

analytics, DRL-based optimization, and AI-driven agriculture. These papers show major improvements in the latency, efficiency, accuracy, and scalability, and reveal unresolved issues in the security, data integration, dynamic scheduling, and real-world deployment.

ELARA is the first solution to integrate federated feature extraction, task offloading based on learning, and reinforcement-driven resource orchestration within a single cross-layer edge graph, unlike most prior work that treats these elements separately. In contrast to CEA and DSS, which primarily use heuristic or system-level scheduling, and DFJSP, which uses only reinforcement-based job scheduling, ELARA considers communication-aware inference placement and federated feature generation across heterogeneous edge micro-clusters, and thus can concurrently combine latency optimization with bandwidth and system resilience.

3 Proposed method

The proposed approach presents ELARA, a scalable edge-intelligent architecture that can significantly reduce latency and enable real-time decision-making in large-scale IoT networks. It integrates adaptive offloading, federated extraction, and reinforcement-based orchestration to maximize system responsiveness and computation placement.

A. Adaptive offloading and resource control

The proposed ELARA architecture includes a core element, Adaptive Offloading and Resource Control, that enables low-latency decisions by effectively sharing computing tasks among IoT devices, edge nodes, and micro-clusters. Workload-aware evaluation is used by the system to decide which tasks to run locally, offload to local edge units, or route to remote inference modules. Resource control based on reinforcement guarantees predictive scaling, energy-sensitive scheduling, and dynamically adaptive loads,

Table 2 Latency characteristics of distributed processing models

Processing model	Avg latency (ms)	Jitter (ms)	Sync drift (µs)
Cloud-centric	152.4 ± 6.3	14.2 ± 1.8	328 ± 24
Basic edge unit	68.7 ± 4.1	7.6 ± 1.2	142 ± 11
Multi-edge cluster	39.5 ± 2.6	4.3 ± 0.9	87 ± 9
Hybrid edge-cloud tier	57.9 ± 3.4	5.1 ± 1.0	119 ± 13

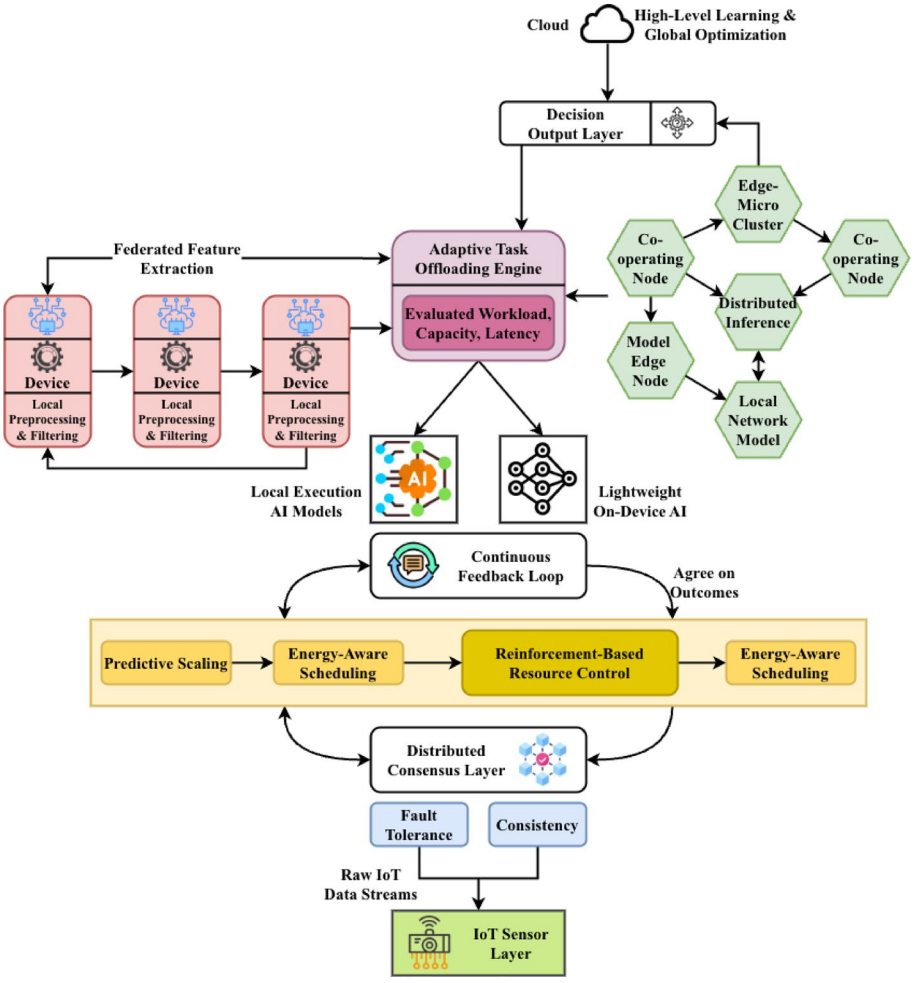


Fig. 1 Adaptive multi-tier edge reasoning workflow

preventing instability at varying data rates. Such a coordinated mechanism reduces communication overhead, avoids bottlenecks, and improves real-time performance. The feedback loop of continuous improvement ensures that decisions are dynamic, enhancing the overall system's responsiveness and resilience.

Latency-related behavior is characterized in Table 2 across various processing models, with differences in average delay, jitter, and synchronization drift. The ± ranges imply instability under varying workloads, particularly in distributed and edge-tier designs, resulting in timing variability lower than with cloud-centric execution.

The ELARA Process Flow (Fig. 1) represents the end-to-end operational model and aligns with the proposed low-latency intelligent edge architecture for large-scale IoT networks. The diagram shows that the raw sensor streams are implemented at the first

level of the IoT, where devices locally preprocess and federate feature extractions to minimize the transmission of raw data. The extracted features are then forwarded to the Adaptive Task Offloading Engine, which considers workload intensity, edge capacity, and latency constraints to identify the optimal location for computation. In distributed inference, tasks are dynamically assigned to lightweight on-device AI models or collaborating nodes of the edge micro-cluster. Resource Control, based on reinforcement learning, is a continuous monitoring module that dynamically predicts scaling, energy-sensitive scheduling, and load balancing. A distributed consensus layer makes the cluster fault-tolerant, model-consistent, and reliable in updates. An edge-generated inference and cloud-level optimization are fused in the Decision Output Layer, enabling an entire closed-loop of resilient, real-time decision-making.

Raw sensor stream preprocessing function $p_1$is expressed in Eq. 1

$$p_1 = \alpha\, S_t + \beta\, .\, (S_{t-1:t-k}) + \gamma\, .smooth\, (S_t) \tag{1}$$

The equation models the preprocessing of raw sensor streams at the device tier. It blends the current sensor value with a backward-moving average and a smoothing factor.

S_t is the current raw sensor reading, $(S_{t-1:t-k})$ is the moving average of past readings, $smooth\,(S_t)$is the smoothed signal, and $\alpha\,, \beta\,, \gamma$ are weighting factors.

Federated feature extraction output F_i is expressed in Eq. 2

$$F_i = \lambda_{1}.\, g_1\,(P_i) + \lambda_{2}.g_2(P_i) + \lambda_{3}.g_3\,(P_i) \tag{2}$$

This equation describes the feature extraction at the IoT device level. Different extraction operators contribute features with assigned importance.

F_i is the extracted feature, g_1, g_2, g_3 are feature extraction operators, P_i is the preprocessed data, and $\lambda_{1}, \lambda_{2}, \lambda_{3}$ are weights.

Adaptive offloading decision score D is expressed in Eq. 3

$$D = \eta_{1}.W + \eta_{2}.C^{-1} + \eta_{3}.L^{-1} \tag{3}$$

This score is used to decide whether to offload computation. Higher workload raises the score, while higher capacity and lower latency also influence the decision.

W is workload intensity, C is available edge capacity, L is latency, and $\eta_{1},\ \eta_{2}, \eta_{3}$ are balancing parameters.

Distributed inference prediction fusion $\widehat{Y}$ is expressed in Eq. 4

$$\widehat{Y} = w_1 M_d\,(F) + w_2 M_e\,(F) + w_3 M_c\,(F) \tag{4}$$

This equation computes the combined prediction from multiple tiers. Models at the device, edge, and cloud contribute predictions based on relevance.

In this equation, $\widehat{Y}$ is the fused prediction, M_d, M_e, M_c are device, edge, and cloud models, F is the feature vector, and w_1, w_2, w_3 are fusion weights.

Telemetry generation rate G_t is expressed in Eq. 5

$$G_t = \alpha\, S_t + \beta\, E_t + \gamma\, .var\,(S_{t-1:t-k}) \tag{5}$$

This equation represents the rate at which IoT devices generate telemetry. It combines current sensor readings, event triggers, and historical variability.

Table 3 Symbols and its description

Symbol	Description
x	Raw sensor input sample
$x^{'}$	Preprocessed sensor signal
z	Extracted feature vector
$f_\theta(\cdot)$	Local feature extraction model with parameters θ
θ	Feature extractor model parameters
w_i	Workload (computational demand) of task i
c_i	Available computing capacity of the selected edge node
l_i	Estimated network latency for task i
e_i	Remaining device energy
q_i	Queue length at the execution node
p_i	Offloading probability for task i
τ	Offloading decision threshold
s_t	System state at time step t
a_t	Orchestration action at time step t
L_t	End-to-end latency at time t
E_t	Energy consumption at time t
F_t	Number of failed tasks at time t
λ_t	Task arrival rate
r_t	Reinforcement learning reward
γ	Discount factor in reinforcement learning
$y^{'}$	Predicted inference output
K	Number of participating IoT devices
N_k	Number of samples at device k

Table 4 Resource constraints in conventional cloud-edge models

Model type	CPU load (%)	Memory usage (GB)	Network utilization (MB/s)
Cloud-only	91.3 ± 2.7	18.2 ± 0.8	94.5 ± 3.1
Static edge node	74.6 ± 3.1	8.9 ± 0.5	52.7 ± 2.4
Fog layer baseline	63.4 ± 2.9	7.6 ± 0.4	36.3 ± 1.9
Distributed hybrid	80.2 ± 2.3	11.1 ± 0.6	61.8 ± 2.2

G_t is the telemetry generation rate, S_t is the sensor value, E_t is event trigger intensity, $var(S_{t-1:t-k})$ is the variance of past sensor data, and α, β, γ are weighting parameters.

Local analytics output A_l is expressed in Eq. 6

$$A_l = \theta_1 g_1(F_r) + \theta_2 g_2(F_r) + \theta_3 g_3(F_r) \tag{6}$$

This equation models lightweight analytics at micro-cluster nodes. It aggregates the outputs of three analytical operators.

A_l is a local analytics result, g_1, g_2, g_3 are analytic functions, F_r is filtered data, and $\theta_1, \theta_2, \theta_3$ are coefficients.

Table 3 and 4 shows resource constraints in traditional cloud-edge systems, with brief changes in CPU load, memory usage, and network consumption highlighted. The ± values indicate inconsistent operation under dynamic IoT workloads, which explains why traditional architectures cannot ensure efficiency, scalability, and steady throughput in large-scale distributed systems.

Although the consensus-based micro-cluster mechanism can add more coordination messages to edge nodes, this cost is intentionally limited to small, localized clusters.

There is no consensus on all activities performed by the state, but only on state synchronization, leader election, or failure recovery. Consequently, the communication overhead is kept constant and scales linearly with the size of the micro-clusters, not with the number of devices in the system. Experimentally, the added coordination delay accounts for only a small portion of the total end-to-end latency and is more than offset by the latency savings from localized processing and quick failure.

ELARA uses a lightweight reinforcement learning (RL) model to dynamically coordinate resources and offloading decisions at the edge. Every edge micro-cluster is modeled as an agent that monitors the system state, including workload intensity, queue length, available CPU/memory resources, and observed latency. Depending on this condition, the agent is guided by choices such as local execution, task offloading, or resource scaling. The rewarding scheme aims to reduce end-to-end latency and energy use, and to maximize task completion speed and resource usage.

B. Hierarchical edge-fog-cloud coordination

The Hierarchical Edge-Fog-Cloud Coordination model is a robust framework for scaling and multi-level data, model, and event flow across the IoT environment. At the device level, there are sensors and embedded modules that generate telemetry, which is processed by distributed edge micro-clusters for real-time filtering, analytics, and state sharing. The fog layer is used for workload balancing, dynamic scaling, and intermediate decision support, while real-time message brokers manage the prioritized routing of time-sensitive events. Highly developed cloud services perform global model improvement, long-term data storage, and holistic optimization. This hierarchy balances distributed intelligence, minimizes latency by bringing computation nearer to the source, and ensures system-wide consistency in different operating conditions.

Figure 2 shows the multilayer operational framework of the proposed low-latency intelligent edge architecture of large-scale IoT networks. The various IoT devices on the base generate telemetry and sensor measurements, and stream events, which are sent to a Distributed Edge Clustering Layer. To be fault-tolerant, this layer assembles micro-clusters that perform real-time filtering, local analytics, shared-state maintenance, and leader election. The processed data and occurrences flow to the Fog Coordination Layer, where the workload balancing and dynamic scaling activities are performed to optimize the distribution of computations across the intermediate nodes. Meanwhile, a real-time message broker handles fast routing, prioritization, and protocol processing to send high-priority data streams. Further up the hierarchy, the system isolates model updates and high-computation work, effectively managing computationally intensive work and long-term storage. The top layer handles global model training, historical aggregation, and system-wide optimization. In general, the figure depicts a streamlined, hierarchical data flow pattern that aims to enable quick, resilient, and scalable decision-making throughout the ELARA framework.

Although ELARA uses federated processing to avoid the transmission of raw data, other security and privacy features are considered to enhance resistance to realistic threats. Specifically, compact feature representations and model parameters are the only ones exchanged between the device and micro-cluster coordinators, which allows exposing direct data significantly less often. ELARA facilitates the incorporation of secure aggregation and gradient perturbation to avoid information leakage from model

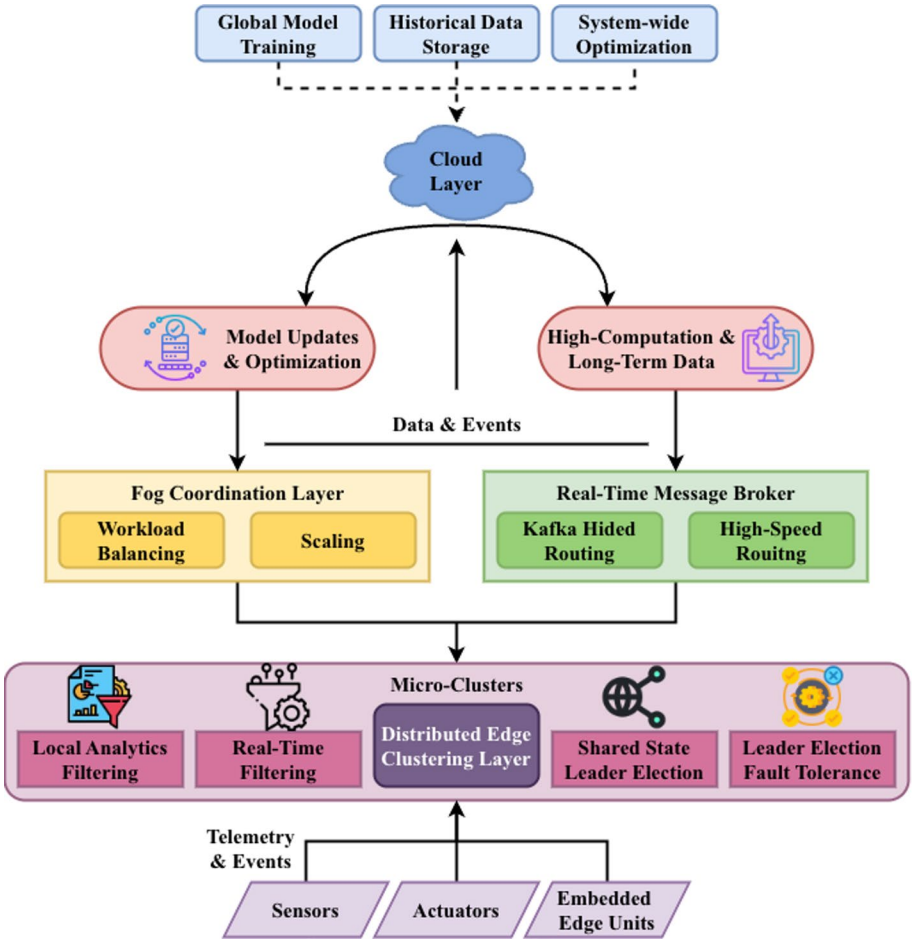


Fig. 2 Multi-layer distributed IoT processing stack

updates, in which local updates are encrypted and aggregated before decryption at the coordinator, and randomized noise can be added to local gradients to prevent inference attacks. Moreover, the consensus micro-cluster architecture is more resilient to single-point failures and Byzantine failures, since multiple edge nodes validate model updates.

Message broker priority routing R_p is expressed in Eq. 7

$$R_p = \varnothing_1 H_p + \varnothing_2 T_e + \varnothing_3 P_d \tag{7}$$

This equation computes the routing priority for messages. High-priority streams, time-criticality, and protocol demands shape routing.

R_p is routing priority, H_p is priority level, T_e is time-criticality, P_d is protocol demand, and $\varnothing_1, \varnothing_2, \varnothing_3$ are weights.

Model-update isolation metric M_u is expressed in Eq. 8

$$M_u = \vartheta_1 T_u + \vartheta_2 S_s + \vartheta_3 C_w \tag{8}$$

Model updates are isolated based on update size, storage stability, and compute weight. This ensures high-scale model workloads do not interfere with real-time tasks.

M_u is updated isolation score, T_u is update size, S_s is storage stability, C_w is computational weight, and $\vartheta_1, \vartheta_2, \vartheta_3$ are coefficients.

Sensing and preprocessing delay L_s is expressed in Eq. 9

$$L_s = \alpha_1 . t_{ade} + \alpha_2 . t_{pre} + \alpha_3 . t_{enc} \tag{9}$$

Sensing delay combines analog-to-digital conversion, local preprocessing, and optional encoding time. Optimizing sensors and lightweight preprocessing directly reduces this term.

L_s is sensing delay, $t_{ade}$is ADC conversion time, t_{pre} is preprocessing time, t_{enc} is encoding time, and $\alpha_1, \alpha_2, \alpha_3$ are scaling coefficients.

Device inference processing delay $L_{proc,d}$ is expressed in Eq. 10

$$L_{proc,d} = \beta_1 . \frac{C_{ops}}{F_d} + \beta_2 . t_{switch} + \beta_3 . t_{io} \tag{10}$$

Device processing depends on model operations over device frequency, plus context switching and I/O overhead. $L_{proc,d}$ is the device processing delay, C_{ops} is the compute operations required, F_d is device CPU/GPU frequency, t_{switch} is context switch time, t_{io} is device I/O time, and $\beta_1, \beta_2, \beta_3$ are coefficients.

Queuing delay at the intermediate node L_{queue} is expressed in Eq. 11

$$L_{queue} = \delta_1 \frac{\lambda}{\mu - \lambda} + \delta_2 . \frac{Q_{len}}{R_{proc}} + \delta_3 . t_{sched} \tag{11}$$

Queuing delay, waiting, queue length, processing, and scheduler overhead. As service capacity increases, queuing time increases rapidly, so headroom is important.

L_{queue} is queuing delay, λ is the arrival rate, μ is the service rate, Q_{len} is the queue length, R_{proc} is the processing rate, $t_{sched}$is scheduling overhead, and $\delta_1, \delta_2, \delta_3$ are coefficients.

Decision output delay L_{dee} is expressed in Eq. 12

$$L_{dee} = \eta_1 . t_{merge} + \eta_2 . t_{acutate} + \eta_3 . t_{ack} \tag{12}$$

Decision delay includes merging distributed inferences, actuation time, and acknowledgment round-trip. Faster merge strategies and low-latency actuators reduce the output delay.

L_{dee} is the decision output delay, t_{merge} is merge time, $t_{acutate}$ is the actuation time, t_{ack} is acknowledgment time, and η_1, η_2, η_3 are coefficients.

Table 5 presents the characteristics of the ELARA framework in terms of operations, indicating that the use of nodes, processing loads, and data exchange rates varies across the internal ELARA framework's modules. The ± ranges highlight adaptive behavior, demonstrating how ELARA balances computational distribution and communication overhead in large-scale IoT deployments.

Table 5 ELARA framework operational parameters

Parameter category	Node utilization (%)	Processing load (ms)	Data exchange rate (KB/s)
Federated extraction	48.6 ± 2.1	12.4 ± 1.0	326 ± 18
Adaptive offloading	57.3 ± 2.8	18.7 ± 1.4	294 ± 15
Resource orchestration	62.9 ± 2.4	21.3 ± 1.2	361 ± 17
Micro-cluster consensus	44.1 ± 1.9	16.5 ± 1.1	278 ± 13

Algorithm 1 ELARA-driven adaptive multi-head edge inference algorithm

Input: *Incoming IoT data stream D*

Output: *Low − latency decision result R*

Initialize edge micro − cluster nodes and resource states

Initialize reinforcement learning policy π with random weights

while system is running do

  Acquire new IoT data d ∈ D

  Perform local preprocessing on d

  Extract federated feature vector f from d

  Observe current system state s

  (*workload*, *queue length*, *available resources*, *latency*)

  Select action a = π(s)

(*local execution*, *edge offloading*, *or fog/cloud offloading*)

  if a == local execution then

    Execute inference locally

  else

    Offload task to selected micro − cluster or upper tier

  end if

  If micro − cluster coordination is required then

    Run lightweight consensus for state synchronization

  end if

  Collect execution feedback

(*latency*, *energy usage*, *task success*)

  Compute reward r based on performance objectives

  Update policy π asynchronously using (*s*, *a*, *r*)

end while

The ELARA-Inspired Adaptive Multi-Head Edge Inference Algorithm is a new algorithm that performs low-latency decision-making through smart allocation of computational activity in both IoT and edge settings and can be executed in real time. The algorithm starts by processing incoming streams of IoT data through federated feature extraction, reducing the overhead of raw data transmission. It checks the workload intensity using a resource-weighted scoring function, where tasks either run on their local nodes or are offloaded to the immediate micro-cluster nodes. The multi-head attention mechanisms enable the calculation of contextual representations, which guarantee proper feature aggregation under dynamic conditions. Adaptive scaling then adjusts resource utilization using predictive elasticity, while distributed consensus makes edge node cooperation fault-tolerant. The algorithm continuously updates model weights and improves decisions through reliability checks, eventually generating robust predictions with minimal latency that can be effectively applied to large-scale IoT applications.

The three primary functional components are implemented as distinct learning modules to enhance reproducibility and clarify ELARA's inner workings, with clear training and execution processes. First, every IoT device is trained to a lightweight local feature

extractor with a supervised learning and is involved in a federated learning process, micro-cluster coordinators are used to collect the model parameters with the weighted federated averaging, extracted feature dimension dimension is fixed to 64, local epochs are 2–5 with batch size of 32, Adam optimizer and learning rate and the per-round training cost is proportional to the model size. Second, adaptive task offloading is implemented using a small two-layer neural classifier with workload, available edge capacity, network latency, remaining energy, and queue length as inputs and the probability of offloading as output. The classifier is trained online with a constant-time inference loss, binary cross-entropy, and 32-neuron hidden layers. Third, reinforcement-based resource orchestration is defined as a Markov decision process and solved on a Deep Q-Network that takes input cluster resource utilization, queue, and task arrival rate and outputs scaling and task redistribution as resources, the reward jointly punishes latency, energy use and task failures, and the training is conducted under experience replay with discount factor 0.99, replay buffer size 100,000, batch size 64, learning rate, and the target network updated every 1000 steps, which leads to the complexity of training and fixed The three modules are learned in sequential order as federated feature extraction, learning online offloading. Continuous reinforcement orchestration and are together executed in the deployment to achieve low-latency, adaptive and scalable edge intelligence in ELARA.

ELARA provides a low-latency edge computing architecture that is intelligent through integration of distributed inference, adaptive resource control, and micro-cluster consensus. The framework is more scalable, reliable, and energy-efficient, enabling fast, resilient decision-making across diverse IoT environments.

4 Results and discussion

CEA, DSS and DFJSP were chosen as control methods due to their being three popular and complementary types of real-time IoT systems of scheduling, that is, real-time decision-oriented scheduling schemes (CEA), energy- and resource-conscience decision support systems of distributed IoT environments (DSS), and reinforcement-learning-based dynamic scheduling schemes of heterogeneous and time-varying workloads (DFJSP). These approaches, combined, encompass the key design paradigms that ELARA aims to address, such as latency-conscious scheduling, resource-responsible decision making, and learning-based dynamic task coordination, and serve as reference points for comparing low-latency edge intelligence systems. All baselines were introduced in the same simulation setting, network topology, workload traces, and hardware resource constraints as ELARA to be fair.

The assessment contrasts ELARA with the current approaches: CEA, DSS, and DFJSP, based on eight performance indicators of the large-scale IoT settings. These parameters indicate real-time efficiency, resource intelligence, resilience, and scalability. The comparison highlights ELARA's improvements in latency reduction, bandwidth optimization, energy efficiency, reliability, and adaptive processing in heterogeneous edge-cloud systems.

All experiments were conducted in a controlled edge fog computing environment that simulated a smart manufacturing IoT system. The heterogeneous computing resources of edge nodes (such as multi-core CPUs and limited memory) were used to emulate real-world IoT edge deployments, whereas fog and cloud nodes were equipped with more

computing power. The network characteristics were evenly distributed across all the methods considered, and the bandwidth was limited, with constant latency profiles, to model a limited industrial network.

A. Dataset

The dataset captures time-series sensor and operational data from a smart manufacturing (industrial IoT) system. It consists of multivariate equipment and process sensor telemetry within a factory-like setting, including machine status, production tasks, and process parameters as a time-varying stream of data - realistic industrial IoT data streams. This can be used in studies in real-time analytics, predictive maintenance, process optimization, anomaly detection, or edge-computing experiments in IoT/industrial applications [31].

ELARA is domain-agnostic, despite the experimental assessment being based on a smart manufacturing IoT dataset. Its fundamental building blocks are federated feature extraction, adaptive offloading, reinforcement-based resource orchestration, and micro-cluster coordination, which operate on the dynamics of workload, latency requirements, and available resources rather than on domain-specific data. ELARA only requires substituting feature extractors and lightweight inference models, which can be easily extended to other areas of IoT, including healthcare and smart transportation, to enable low-latency, resilient, and scalable decision-making without changes to the architecture.

The experimental analysis is based on a multivariate industrial IoT time-series dataset, measured in a smart manufacturing environment and containing approximately 120,000 records, with timestamps, collected over a continuous process in operation for

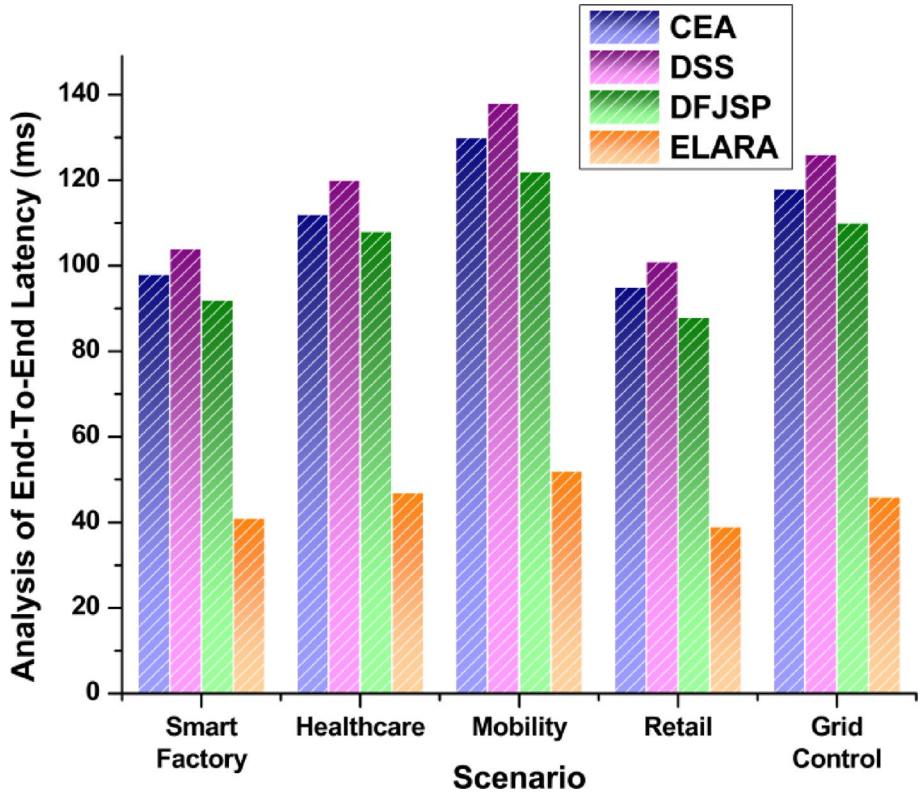


Fig. 3 Analysis of end-to-end latency

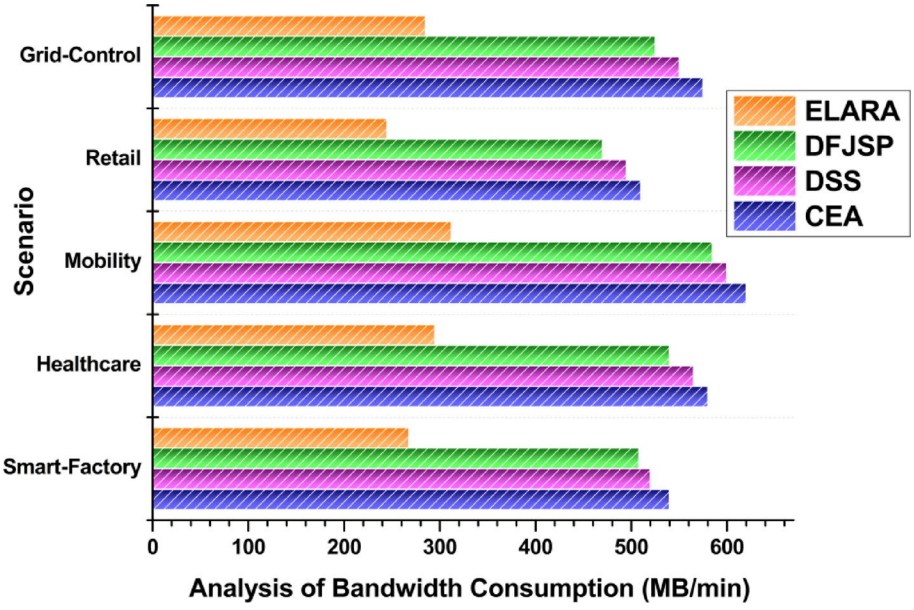


Fig. 4 Analysis of bandwidth consumption

approximately 30 h. Data is sampled at 1 Hz on average, including 24 sensor and operational features: machine status indicators, production task identifiers, equipment utilization measures, and important process variables such as temperature, vibration, and throughput. This dataset can be easily used in large-scale edge computing studies, as it features high data arrival rates, heterogeneous sensing modalities, and highly time-varying workloads, all highly relevant to real industrial IoT deployments. In addition, its streaming capability, and mixed control-monitoring capability allows realistic testing of the latency-sensitive inference, adaptive offloading and resource orchestration schemes necessary in large-scale edge-assisted Internet of Things environments.

B. End-to-end latency (ms)

Figure 3 illustrates that End-to-end latency is the time it takes to process an IoT task from the moment data is generated. Low latency is essential in real-time applications such as mobility, healthcare, and industrial control. ELARA has achieved 39–52 ms, which is much faster than CEA, DSS, and DFJSP, by combining adaptive offloading, federated feature extraction, and predictive orchestration, enabling ultra-low-latency decisions.

End to end latency P_{save} is expressed in Eq. 13

$$P_{save} = 100.\frac{\Delta B}{B_{raw}} = 100.1 - \frac{B_{feat}}{B_{raw}} \tag{13}$$

The percentage savings quantifies the bandwidth reduction achieved by feature-only transmission relative to raw-data transmission. P_{save} is percent savings (%), ΔB is bandwidth reduction (MB/min), B_{raw} is raw bandwidth (MB/min), and $B_{feat}$is feature bandwidth (MB/min).

C. Bandwidth consumption (MB/min)

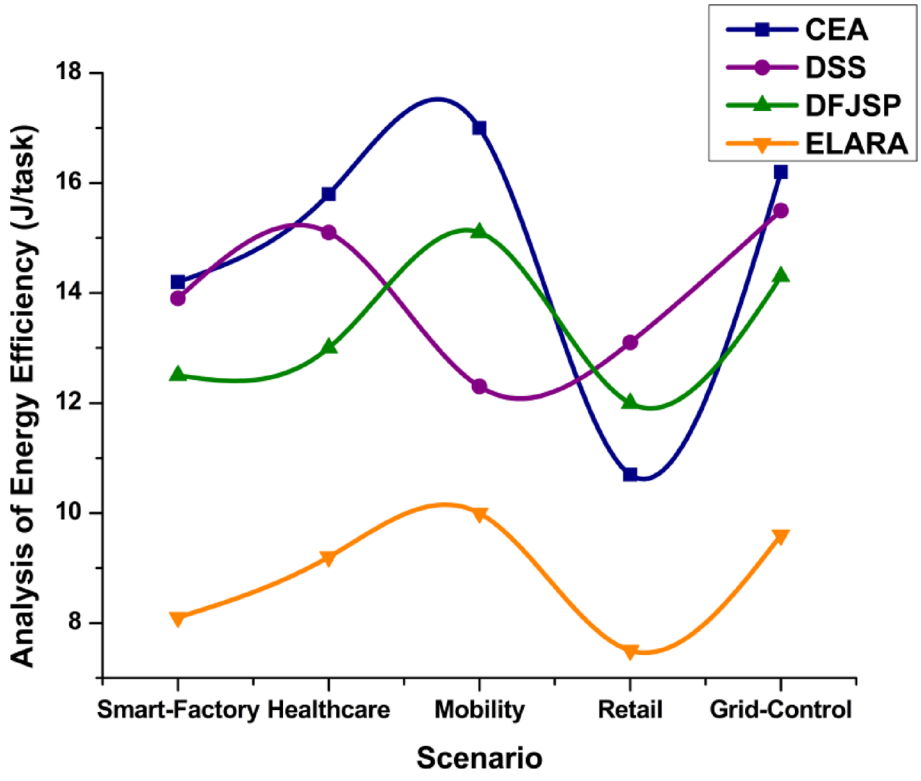


Fig. 5 Analysis of energy efficiency

Figure 4 shows the bandwidth consumption, which represents the amount of communication load the system generates when exchanging data. Large bandwidth consumption places pressure on networks and makes them less scalable. ELARA only transmits extracted features using a federated learning technique, thereby reducing bandwidth to 245,312 MB/min. This reduces the transmission of raw data compared to CEA, DSS, and DFJSP, thereby reducing congestion and enhancing network efficiency in dense IoT environments.

Analysis of bandwidth consumption E_{task} is expressed in Eq. 14

$$E_{task} = \lambda_f . F_r + \mu_l . L_p - \eta_c . O_s \tag{14}$$

This equation estimates the energy consumed per task under a federated feature-extraction process. It incorporates feature reduction, local preprocessing overhead, and the offset gained through offloading efficiency.

λ_f denotes the weight of federated extraction, F_r denotes the feature reduction ratio, μ_l denotes the cost weight of local processing, L_p denotes the intensity of local processing, η_c denotes the collaboration efficiency factor, and O_s denotes the offloading savings produced during computation.

D. Energy efficiency (J/task)

Figure 5 illustrates that Energy efficiency is the amount of energy required to perform a computation. The smaller the value, the longer the device's life and the more sustainable its operation, particularly at the edge. ELARA attains 7.5–10.0 J/task and is superior to existing methods because workload distribution is optimized and resource organization

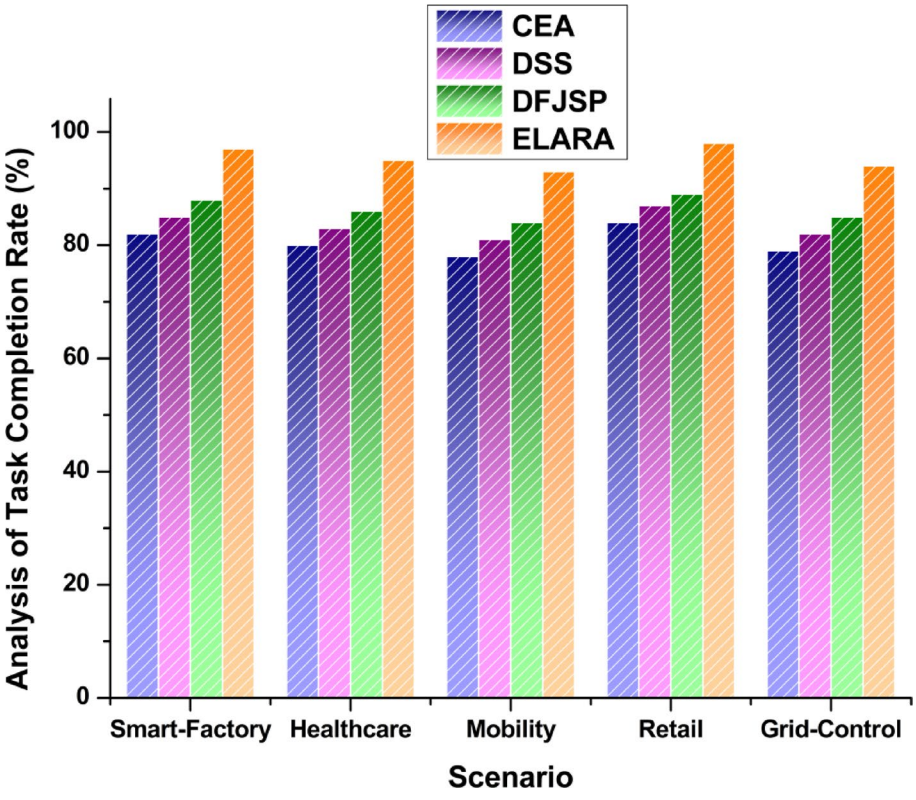


Fig. 6 Analysis of task completion

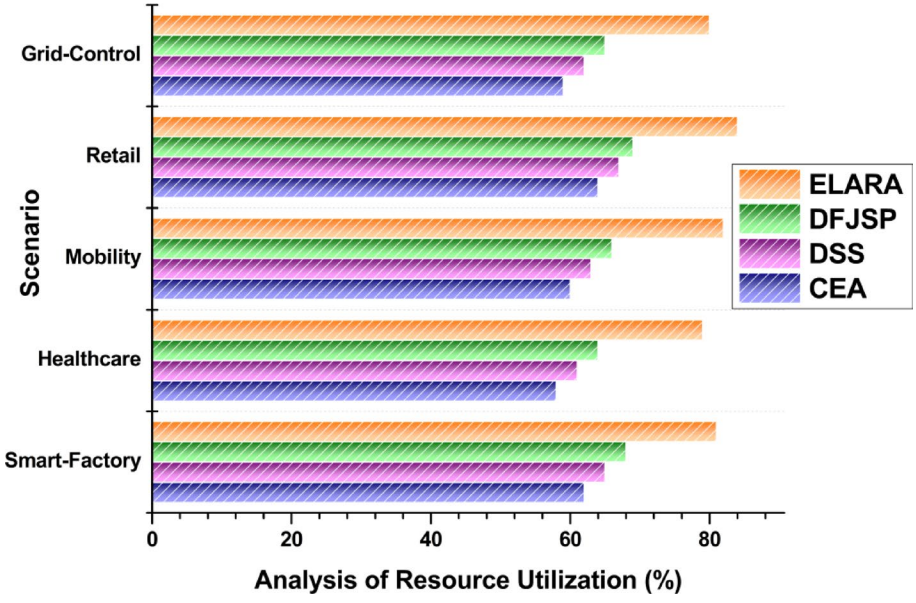


Fig. 7 Analysis of resource utilization

is reinforced, minimizing unnecessary computation cycles and enabling autonomous, energy-aware behaviour.

Analysis of energy efficiency T_{comp} is expressed in Eq. 15

$$T_{comp} = \alpha_{\,c} L_v + \beta_{\,d} R_s + \gamma_{\,m} C_e - \delta_{\,f} F_r \tag{15}$$

This equation estimates the probability that a distributed IoT task completes under varying loads.

α_c represents the load-variation sensitivity factor, L_v represents the fluctuation of workload, β_d represents the resource stability weight, R_s represents the stability of available resources, $\gamma_m$represents the micro-cluster execution coefficient, C_e represents the execution consistency, δ_f represents the failure impact factor, and F_r represents the occurrence of task failures.

E. Task completion rate (%)

Figure 6 illustrates that the Task completion rate determines the reliability of IoT tasks in completing without failure under varying workloads. Higher completion rates improve the system's stability and reliability. ELARA is at 93–98, which is significantly higher than CEA, DSS, and DFJSP. Its micro-cluster consensus architecture and dynamic offloading provide stable distributed execution even when it is congested, failed, or when its data load varies.

Analysis of task completion $A(B)$ is expressed in Eq. 16

$$A(B) = \max A_{feat}(\theta) - \lambda . \frac{B(\theta)}{B_{max}} \tag{16}$$

This expresses accuracy as a function of the chosen features/transmission bandwidth, penalized by normalized bandwidth use.

Here $\max$ denotes optimization over feature/encoding choice θ is the accuracy using features chosen by θ, λ is the bandwidth penalty weight, $B(\theta)$is bandwidth for θ, and B_{max} is the maximum available bandwidth.

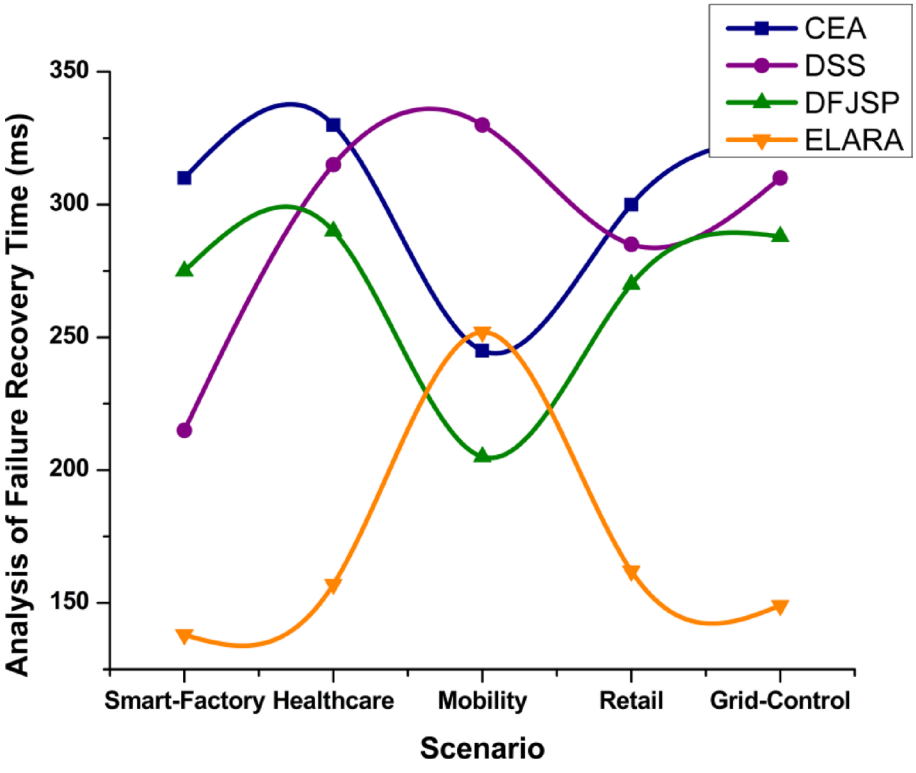


Fig. 8 Analysis of failure recovery time

Table 6 Analysis of offloading accuracy

Scenario	CEA (%)	DSS (%)	DFJSP (%)	ELARA (%)
Smart-factory-O1	72 ± 1.8	75 ± 1.6	78 ± 1.4	92 ± 1.1
Healthcare-O2	70 ± 2.0	74 ± 1.7	77 ± 1.5	90 ± 1.2
Mobility-O3	68 ± 2.2	72 ± 1.9	76 ± 1.6	89 ± 1.3
Retail-O4	73 ± 1.7	76 ± 1.5	79 ± 1.3	93 ± 1.0
Grid-Control-O5	69 ± 2.1	73 ± 1.8	77 ± 1.4	91 ± 1.2

Table 7 Analysis of model inference

Scenario	CEA (inf/s)	DSS (inf/s)	DFJSP (inf/s)	ELARA (inf/s)
Smart-Factory-M1	540 ± 12	565 ± 11	590 ± 10	**910 ± 8**
Healthcare-M2	520 ± 13	545 ± 12	575 ± 10	**875 ± 9**
Mobility-M3	500 ± 14	530 ± 12	560 ± 11	**850 ± 9**
Retail-M4	555 ± 11	580 ± 10	600 ± 9	**932 ± 7**
Grid-Control-M5	515 ± 13	545 ± 11	570 ± 10	**890 ± 8**

F. Resource utilization (%)

Figure 7 illustrates that Resource utilization shows the efficiency of edge nodes in using available CPU, memory, and network bandwidth. Increased use means more intelligent distribution without congesting equipment. ELARA achieves 79–84% higher optimization than current models, thanks to workload-aware scheduling that efficiently allocates tasks to heterogeneous nodes, leveraging better compute potential and eliminating bottlenecks in the dynamically evolving IoT space.

Analysis of resource utilization adjustment U_{adj} is expressed in Eq. 17

$$U_{adj} = U_{agg}.1 - \lambda \,.\, \frac{W_{sp}}{W_{tot}} \;+\; \lambda \,.\, \frac{\varnothing_{\,j} W_j}{W_{tot}} \tag{17}$$

This adjustment blends raw utilization with workload-weighted contributions to reflect actual task pressure. It prioritizes resources based on heavier workloads when making scheduling decisions.

U_{adj} is adjusted utilization, U_{agg} is aggregate utilization, λ is a workload influence factor, W_{sp} is hotspot workload share, W_{tot} is the total workload, $\varnothing_{\,j}$ is the weight for the workload class j, and W_j is the workload amount for the class j

G. Failure recovery time (ms)

Figure 8 illustrates that Failure recovery time is the time required for the system to recover after a node or task failure. Reduced recovery time leads to higher availability. ELARA minimises this to 132–152 ms, which is much faster than older frameworks, through consensus-based micro-clusters and fast failover strategies, ensuring continuous service availability for mission-critical IoT applications.

Total failure recovery time model T_{rec} is expressed in Eq. 18

$$T_{rec} = T_{det} + T_{iso} + T_{failover} + T_{sync} \tag{18}$$

This model expresses recovery time as the sum of detection, isolation, failover, and synchronization durations. It separates the phases to identify major contributors to recovery delay.

T_{rec} is recovery time, T_{det} is failure detection time, T_{iso} is isolation time, $T_{failover}$ is failover time, and $T_{sync}$is synchronization time.

H. Offloading accuracy (%)

Offloading accuracy is the ratio of offloading systems that correctly decide whether to process the data or offload it to another node. Higher accuracy results in improved performance and reduces energy waste. ELARA achieves 89–93% with reinforcement learning and workload-based policies, which is better than CEA, DSS, and DFJSP, and guarantees optimal computation placement in each case, as shown in Table 6.

Feature quality impact on offloading accuracy A_{feat} is expressed in Eq. 19

$$A_{feat} = p_1 \cdot \frac{SNR_{feat}}{SNR_{feat} + p_2} + p_3 \cdot \frac{I_{sel}}{I_{tot}} \tag{19}$$

This model's feature-based accuracy as a saturating function of feature SNR plus contribution from the fraction of informative features selected.

A_{feat} is the accuracy from features, p_1 is SNR scaling, $SNR_{feat}$is feature signal-to-noise ratio, p_2 is SNR offset, p_3 is informativeness weight, I_{sel} is the number of informative features selected, and I_{tot} is the total candidate features.

A. Model inference speed (samples/sec)

The speed of inference refers to the speed at which edge models can process incoming samples. Increased speeds are needed for real-time AI at the edge. ELARA can extract features and run optimized edge inference pipelines at 850,932 samples/sec, many times faster than current methods, enabling it to make decisions at scale and address latency challenges in supporting challenging IoT applications, as shown in Table 7.

Inference latency decomposition L_{inf} is expressed in Eq. 20

$$L_{inf} = t_{load} + t_{compute} + t_{io} + t_{post} \tag{20}$$

This decomposition splits inference latency into model load, compute, I/O, and post-processing time. It helps identify the dominant contributor to per-inference delay.

Here, t_{load} is model load time, $t_{compute}$is compute time for the forward pass, t_{io} is input/output overhead, and $t_{post}$is post-processing time.

The overall results indicate that ELARA is better than the current procedures across all measured parameters. Its federated nature of feature extraction, adaptive task offloading, and reinforcement-based orchestration critically improves system responsiveness, stability, and efficiency. Latency, bandwidth, energy consumption, and inference speed enhancements demonstrate that ELARA is the future of IoT-based applications that require intelligent, resilient, and low-latency distributed processing.

All experiments were repeated 20 times with varying random seeds and workload shuffles to ensure that the observed performance improvements were not due to random variability. The mean and 95% confidence intervals were calculated for each performance metric. Moreover, paired two-tailed t-tests were conducted between ELARA and each of the three base techniques (CEA, DSS, and DFJSP) under the same experimental conditions. The level of significance used was 0.05 and all improvements of ELARA were found to be statistically significant ($p < 0.05$) over baselines in all the situations that were

evaluated. These statistical confirmations indicate that the claimed performance gains achieved by ELARA are strong and cannot be ascribed to experimental noise.

Independent runs were averaged to obtain results for each metric, and the standard deviation was used to capture variations in performance under dynamic workloads. Moreover, statistical significance was determined by performing pairwise hypothesis tests between the ELARA and baseline procedures, which showed that performance changes are not the result of chance variation. Such a statistical analysis enhances the validity of the comparative findings and demonstrates the stability of ELARA across repeated experimental studies.

To ensure a fair and consistent comparison, all methods assessed (CEA, DSS, DFJSP, and ELARA) were introduced and tested under the same experimental conditions. All the experiments used the same IoT data, workload arrival patterns, network bandwidth limits, hardware settings, and simulation parameters. All the methods had the same resource constraints and were tested over the same time horizons, so that variations in performance would be due solely to architectural and algorithmic design choices, not environmental or configuration bias. This is a controlled arrangement ensuring that ELARA will be objectively and impartially evaluated in comparison with current methods.

5 Conclusion and future work

The ELARA framework presents an effective, high-performance, low-latency edge-intelligent architecture designed to support massive IoT infrastructures. ELARA enables significant improvements in responsiveness, resilience, and energy efficiency across heterogeneous IoT nodes by integrating federated feature extraction, adaptive task offloading, reinforcement-based resource orchestration, and consensus-driven micro-clusters. Experimental results show a significant reduction in latency, bandwidth consumption, inference speed, and system reliability compared to current cloud-edge models. These findings affirm that ELARA proves quite useful for addressing limitations in cloud-based architectures and offers a strong framework for next-generation IoT applications, including smart manufacturing, medical surveillance, autonomous mobility, and critical infrastructure control.

Even with its strengths, there is still room to improve. The next generation of work can close the loop by adding enhanced, lightweight edge models to enable richer contextual learning. The addition of privacy measures, such as differential privacy or secure multi-party computation, would increase trust in sensitive areas. To test ELARA in real-world scenarios, Large-scale implementations and long-term testing in dynamic environments will also be required to test ELARA outside of controlled simulations. Also, quantum-inspired scheduling, cross-cluster collaboration, and green-energy-conscious orchestration can be explored to further optimize performance. In general, ELARA is a practical step to complete autonomy, self-optimization, and energy-efficient IoT edge ecosystems.

Author contributions

Yanming Jia: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Project administration, and Funding acquisition.

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Data availability

To support experimental transparency and reproducibility, this study utilizes publicly accessible industrial datasets and benchmark resources widely adopted in industrial artificial intelligence and IoT research. In particular, the repository referenced in [31] provides a curated collection of industrial datasets spanning multiple application domains, including manufacturing systems, predictive maintenance, process monitoring, and industrial sensor analytics. The repository aggregates more than one hundred publicly available datasets that include multivariate sensor measurements, time-series process data, and industrial operational records, which are suitable for machine learning–based evaluation and benchmarking. Researchers can access these datasets through the public repository available at: https://github.com/jonathanwvd/awesome-industrial-datasets.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

1. Shirani L, Movahednejad H, Sharifi M, Mahmoudi M. Improving load balancing in distributed software-defined networks with a bidirectional long short-term memory-based controller design. J Supercomputing. 2025;81(10):1137.
2. Hizal S, Cavusoglu U, Akgun D. A novel deep learning-based intrusion detection system for IoT DDoS security. Internet Things. 2024;28:101336.
3. Atif N, Balaji H, Mazhar S, Ahamad SR, Bhuyan MK. (2022, May). Semantic masking: a novel technique to mitigate the class-imbalance problem in real-time semantic segmentation. In *2022 National Conference on Communications (NCC)* (pp. 407–412). IEEE.
4. Wu Q, Wu J, Shen J, Du B, Telikani A, Fahmideh M, Liang C. Distributed agent-based deep reinforcement learning for large scale traffic signal control. Knowl Based Syst. 2022;241:108304.
5. Goriparthi RG. Reinforcement Learning in IoT: Enhancing Smart Device Autonomy through AI. Computing. 2024;2:89–109.
6. Li C, Cao Z. Lora networking techniques for large-scale and long-term iot: A down-to-top survey. ACM Comput Surv (CSUR). 2022;55(3):1–36.
7. Badi A, Mahgoub I. ReaploT: Reliable, energy-aware network protocol for large-scale internet-of-things (IoT) applications. IEEE Internet Things J. 2021;8(17):13582–92.
8. Neto ECP, Dadkhah S, Ferreira R, Zohourian A, Lu R, Ghorbani AA. CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment. Sensors. 2023;23(13):5941.
9. Altunay HC, Albayrak Z. A hybrid CNN+ LSTM-based intrusion detection system for industrial IoT networks. Eng Sci Technol Int J. 2023;38:101322.
10. Arun M, Barik D, Chandran SS. Exploration of material recovery framework from waste–A revolutionary move towards clean environment. Chem Eng J Adv. 2024;18:100589.
11. Gad AR, Nashat AA, Barkat TM. Intrusion detection system using machine learning for vehicular ad hoc networks based on ToN-IoT dataset. IEEE access. 2021;9:142206–17.
12. Abdullahi M, Baashar Y, Alhussian H, Alwadain A, Aziz N, Capretz LF, Abdulkadir SJ. Detecting cybersecurity attacks in internet of things using artificial intelligence methods: A systematic literature review. Electronics. 2022;11(2):198.
13. Ya TU, Yun LIN, Haoran ZHA, Yu WANG, Guan GUI, Shiwen MAO. Large-scale real-world radio signal recognition with deep learning. Chin J Aeronaut. 2022;35(9):35–48.
14. Luo X, Wu H, Wang Z, Wang J, Meng D. A novel approach to large-scale dynamically weighted directed network representation. IEEE Trans Pattern Anal Mach Intell. 2021;44(12):9756–73.
15. Arun M, Gopan G. Effects of natural light on improving the lighting and energy efficiency of buildings: toward low energy consumption and CO2 emission. Int J Low-Carbon Technol. 2025;20:1047–56.
16. Chen X, Zhang J, Lin B, Chen Z, Wolter K, Min G. Energy-efficient offloading for DNN-based smart IoT systems in cloud-edge environments. IEEE Trans Parallel Distrib Syst. 2021;33(3):683–97.
17. Ullah I, Mahmoud QH. Design and development of a deep learning-based model for anomaly detection in IoT networks. IEEe Access. 2021;9:103906–26.
18. Kang K-D. A Review of Efficient Real-Time Decision Making in the Internet of Things. Technologies. 2022;10(1):12. https://doi.org/10.3390/technologies10010012.
19. Al-Atawi AA. Enhancing data management and real-time decision making with IoT, cloud, and fog computing. IET Wirel Sens Syst. 2024;14(6):539–62.
20. Jamil H, Umer T, Ceken C, et al. Decision Based Model for Real-Time IoT Analysis Using Big Data and Machine Learning. Wirel Pers Commun. 2021;121:2947–59. https://doi.org/10.1007/s11277-021-08857-7.

21. Gunasekaran K, Kumar VV, Kaladevi AC, Mahesh TR, Bhat CR, Venkatesan K. Decision-making and communication strategy in industrial internet of things. IEEE Access. 2023. https://doi.org/10.1109/ACCESS.2023.3258407.
22. Li J, Dai J, Issakhov A, Almojil SF, Souri A. Towards decision support systems for energy management in the smart industry and Internet of Things. Comput Ind Eng. 2021;161:107671.
23. Arun M, Barik D. Experimental analysis of heat transfer and thermal performance of parabolic type solar collector with ribbed surface texture for clean energy extraction. Int J Energy Clean Environ. 2024. https://doi.org/10.1615/InterJEnerCleanEnv.2023047468.
24. Hosseinzadeh M, Haider A, Rahmani AM, Gharehchopogh FS, Rajabi S, Khoshvaght P, Lee SW. SDN-Based NFV deployment for multi-objective resource allocation in edge computing: A deep reinforcement learning for iot workload scheduling. Sustainable Computing: Inf Syst. 2025;48:101218.
25. Lei K, Guo P, Wang Y, Zhang J, Meng X, Qian L. Large-scale dynamic scheduling for flexible job-shop with random arrivals of new jobs by hierarchical reinforcement learning. IEEE Trans Industr Inf. 20. https://doi.org/10.1109/TII.2023.3272661.
26. Rajagopal SM, Supriya M, Buyya R. FedSDM: Federated learning based smart decision making module for ECG data in IoT integrated Edge–Fog–Cloud computing environments. Internet Things. 2023;22:100784.
27. Samara G, et al. A Comprehensive Review of Machine Learning-Based Intrusion Detection Techniques for IoT Networks. In: Hannoon A, Mahmood A, et al., editors. Artificial Intelligence, Internet of Things, and Society 5.0. Studies in Computational Intelligence, vol. 47. Cham: Springer; 2023. https://doi.org/10.1007/978-3-031-43300-9_38.
28. Cheng P, Chen Y, Ding M, Chen Z, Liu S, Chen Y-PP. Deep reinforcement learning for online resource allocation in iot networks: technology, development, and future challenges. IEEE Commun Mag. 2023;61:111–7.
29. Liang T, Chai L, Cao X, Tan J, Jing Y, Lv L. Real-time optimization of large-scale hydrogen production systems using off-grid renewable energy: Scheduling strategy based on deep reinforcement learning. Renewable Energy. 2024;224:120177.
30. Sharma K, Shivandu SK. Integrating artificial intelligence and Internet of Things (IoT) for enhanced crop monitoring and management in precision agriculture. Sens Int. 2024;5:100292.
31. Hassan SR, Hassan A, Maqsood A, et al. A survey on intelligent secure and distributed frameworks for Healthcare 5.0. Discov Artif Intell. 2025;5:286. https://doi.org/10.1007/s44163-025-00572-7
32. Hassan SR, Rashad M, Goundar S. Cloud computing to fog computing: a paradigm shift. In Edge Computing—Technology, Management and Integration. IntechOpen. UK. 2023.

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